

Uhlenbeck's Theorem

①

slice theorem: Def (relative Coulomb gauge): if A and B are connections on E , take gauge transformation g s.t. $D_A^*(gCB) - A = 0$

Now, consider a family of gauge transformations $G_t = \exp(-tX)$ $X \in \mathfrak{g}_E$

then $\frac{d}{dt} \Big|_{t=0} G_t(A) = D_A X$ ($G_t(A) = G_t A G_t^{-1} - dG_t \cdot G_t^{-1}$, $\frac{d}{dt} \Big|_{t=0} (G_t A G_t^{-1}) : G_t = \exp(-tX) \approx 1 - tX + O(t^2)$
 $G_t^{-1} = \exp(tX) \approx 1 + tX + O(t^2)$, $G_t A G_t^{-1} \approx (1 - tX) A (1 + tX) = A + t(A X - X A) + O(t^2)$, so $\frac{d}{dt} \Big|_{t=0} (G_t A G_t^{-1}) = [A, X]$
 $\frac{d}{dt} \Big|_{t=0} (dG_t \cdot G_t^{-1}) : dG_t = d(\exp(-tX)) \approx -t dX + O(t^2)$, $G_t^{-1} \approx 1 + tX + O(t^2)$, $dG_t \cdot G_t^{-1} \approx -t dX + O(t^2)$
 $\therefore \frac{d}{dt} \Big|_{t=0} (dG_t \cdot G_t^{-1}) \approx -dX \quad \therefore \frac{d}{dt} \Big|_{t=0} G_t(A) = [A, X] - (-dX) = D_A X$)

$\therefore \text{Im } D_A$ is infinitesimal gauge ~~transformations~~ $T_A D_A$ $D_A : \{G(A) \mid G \in \mathfrak{g}\}$

$\therefore \int \langle \alpha, D_A X \rangle = \int \langle D_A^* \alpha, X \rangle \quad \therefore \text{Ker } D_A^* = (\text{Im } D_A)^\perp$

$\therefore G(B) - A \in (\text{Im } D_A)^\perp$, the space of connections of relative Coulomb gauge is an orthogonal slice of gauge action.

(slice: $\{A + \alpha \mid D_A^* \alpha = 0\}$ $G(B) - A = \alpha$, $G(B) = A + \alpha \in S_A$, move B to S_A).
 $\{A \mid A \text{ is connection}\}$ $T_A A = \mathcal{D}(G_E) = V_A \oplus H_A = \text{Im } D_A \oplus \text{Ker } D_A^*$ $T_A S_A = \{\alpha \mid D_A^* \alpha = 0\}$
 $\{G(B) - A\}$ $\therefore T_A S_A \perp V_A = \text{Im } D_A$
 $\text{Ker } D_A^* = \text{Im } D_A$

Theorem 1 (slice theorem). Assume $p > \frac{n}{2}$. B sufficiently close to A in L^p_1 , a ~~relative~~ relative Coulomb gauge $g \in \mathfrak{g}_E$, satisfying $D_A^*(gCB) - A = 0$, exists. $\mathfrak{g}_E^p$: space of L^p_2 Sobolev gauge transformations, $\subseteq L^p_2(\text{End } E)$

Pf: Let $B = A + \alpha$. AIM: find g s.t. $G(B)$ satisfies $D_A^*(G(B) - A) = 0$

$$G(B) = G(A + \alpha) G^{-1} = \text{Pref } G \cdot G^{-1} \quad (D_B = \text{Pref } + B, \quad D_A = \text{Pref } + A) \quad D_A G = dG + [A, G]$$

$$G(B) - A = G A G^{-1} - A + G \alpha G^{-1} - \text{Pref } G \cdot G^{-1} = G \alpha G^{-1} - (\text{Pref } G \cdot G^{-1} - G A G^{-1} + A G G^{-1}) = G \alpha G^{-1} - (D_A G) G^{-1}$$

Def: $N: A^p_1 \times \mathfrak{g}_E^p \rightarrow L^p(\text{Im } D_A^*)$

$(\alpha, g) \mapsto D_A^* (-D_A g) G^{-1} + G \alpha G^{-1}$ By theorem in last talk, N is smooth nonlinear map.

$\therefore (p > \frac{n}{2} \quad L^p_2 \hookrightarrow C^0) \therefore$ we can let $G = \exp(X)$, take a linearization of N at $(0, 0)$

(α, g) at $(0, I)$, $G = I + X + O(\|X\|^2)$, $G^{-1} = I - X + O(\|X\|^2)$, $D_A G = D_A(I + X) + \dots = D_A X + \dots$, $D_A G \cdot G^{-1} = (D_A X)(I - X) + \dots = D_A X + O(\|X\|^2)$

$$G \alpha G^{-1} = (I + X) \alpha (I - X) + \dots = \alpha + O(\|X\| \|\alpha\|)$$

$$\therefore N(0, \exp(X)) \approx D_A^* (-D_A X + \alpha)$$

$$\therefore L(\alpha, X) = D_A^* (-D_A X + \alpha) = -D_A^* D_A X + D_A^* \alpha$$

Now, we want to prove $L: L^p(\Omega'(g_E)) \times L^p(g_E) \rightarrow L^p(\text{Im } D_A^*)$ is surjective
 $(\alpha, X) \mapsto D_A^* \alpha - D_A^* D_A X$

(2)

$\therefore D_A^* D_A$ is a self-adjoint elliptic operator $\langle D_A^* D_A \psi, \psi \rangle = \langle D_A \psi, D_A \psi \rangle$, $\sigma(D_A^* D_A)(x, \xi) = |\xi|^2 \cdot \text{Id}$.

$\therefore \text{Im } D_A^* D_A$ is closed, $\ker(D_A^* D_A) = \ker(D_A^* D_A)$ (Fredholm operator)
 $\text{Coker}(D_A^* D_A) = \ker(D_A^* D_A)$

$\therefore \int \langle D_A^* D_A X, X \rangle = \int |D_A X|^2 \quad \therefore \ker(D_A^* D_A) = \ker(D_A)$

$\therefore \int \langle D_A^* \beta, X \rangle = \int \langle \beta, D_A X \rangle \quad \forall \beta$, so $\int \langle D_A^* \beta, X \rangle = 0 \iff \int \langle \beta, D_A X \rangle = 0 \quad \forall \beta, D_A X = 0, X \in \ker D_A$

$\therefore \text{Im } D_A^* \perp \ker D_A = \ker(D_A^* D_A)$

$\therefore D_A^* D_A X = \alpha$ is solvable, $\forall \alpha \in \text{Im } D_A^*$ $(L^2(\Omega^0(g_E)) = \text{Im}(D_A^* D_A) \oplus \ker(D_A^* D_A))$
 $\alpha = D_A^* D_A X \rightarrow X$ exists

$\therefore$ Just take $\alpha = 0$, $-D_A^* D_A X$ will cover all $\text{Im } D_A^*$

$\therefore L$ is surjective, by implicit function theorem (Banach space), $\exists \varepsilon > 0$, s.t. $\forall \|a\|_L < \varepsilon, \exists X \in L^p, N(a, e^X) = 0$

Then take $\sigma = e^X$, $D_A^*(\sigma CB) - A = 0$

$$\begin{cases} E = M \times R^k, F = M \times R^l, \text{ for } s \in \Gamma(E), X \in \Gamma(TM), \nabla_X^E s = X(s) \\ E \otimes F = M \times R^{kl} \quad \therefore \nabla_X(s \otimes t) = (\nabla_X^E s) \otimes t + s \otimes (\nabla_X^F t) \end{cases}$$

(here: $s \in \Gamma(E)$, $s(x) = (x, f(x))$, $f: M \rightarrow R^r$, $d = \text{Dref}: d_X s = (X, X(f))$)

Now, we are going to talk about:

Theorem: Uhlenbeck's Theorem: n -dim cpt manifold M , $H^1(M) = 0$. $\mathbb{R}^n \times M \times R^r \quad d = \text{Dref}$ (product connection)

$\frac{n}{2} < p < n$.

$\forall \varepsilon > 0$, def:

$$A_{1, \varepsilon}^p := \{A \in A_1^p : \|FA\|_{L^{n/2}} \leq \varepsilon\}$$

(A_1^p) space of L^p_k Sobolev connections
 $\nabla_A = \nabla_{\text{ref}} + A$
 $\nabla_{\text{ref}}, A \in L^p_k(\Omega^1(g_E))$ not always smooth

$A_i :=$ path connected component including d .

Uhlenbeck's Theorem: For $\varepsilon(M, r) > 0$ small enough, $\forall A \in A_\varepsilon, \exists \sigma \in \mathcal{G}_2^p$ s.t. $d^* \sigma A = 0$

and $\forall \frac{n}{2} \leq q \leq p, \|\sigma A\|_{L^q} \leq C \|FA\|_{L^2}$
 L^p Sobolev gauge trans $\in L^p_k(\text{End } E)$

Lemma. $\exists \eta > 0, N > 0$, depending on M, r, p , s.t. if $d^* A = 0, \|A\|_{L^N} < \eta$

when $q = p$ then $\|A\|_{L^q} \leq N \|FA\|_{L^2} \quad \forall \frac{n}{2} \leq q \leq p$.

Pf: Elliptic estimate:

$$\|A\|_{L^p} \leq C (\|dA\|_{L^p} + \|d^* A\|_{L^p} + \|A\|_{L^p})$$

$\therefore H^1(M) = 0$

$$\|A\|_{L^p} \leq C_1 (\|dA\|_{L^p} + \|d^* A\|_{L^p})$$

(If not, $\|A\|_{L^q} > C (\|dA\|_{L^q} + \|d^* A\|_{L^q}) \quad \forall C$. Take $C = n$, $\exists \{A_n\}$ s.t. $\|A_n\|_{L^q} = 1, \|dA_n\|_{L^2} + \|d^* A_n\|_{L^2} \rightarrow 0$

$\{A_n\}$ bounded in L^2 , by Rellich cpt thm, $\exists A_\infty \rightarrow A_\infty$ in $L^2, A_n \rightarrow A_\infty$ in L^2 .

$\|dA_\infty\|_{L^2} \leq \liminf \|dA_n\|_{L^2} = 0 \Rightarrow dA_\infty = 0; \|d^* A_\infty\|_{L^2} \leq \liminf \|d^* A_n\|_{L^2} = 0 \Rightarrow d^* A_\infty = 0 \therefore A_\infty$ is harmonic 1-form

By Hodge theorem, $A_\infty = 0$. But $\|A_n\|_{L^q} = 1 \quad A_n \rightarrow A_\infty$ in $L^2, L^2 \subset L^q \Rightarrow$ contradiction!

$$H^1_{\text{dR}}(M) \cong H^1(M) : \Delta A = (d d^* + d^* d) A = 0$$

∴ Sobolev embedding $\|A\|_{L^{p'}} \leq C_2 \|A\|_{L^p}$ $1 - \frac{n}{p} = -\frac{n}{p'}$ ($L^p \hookrightarrow L^{p'}$ if $1 - \frac{n}{p} \geq -\frac{n}{p'}$)

Hölder inequality $\|A\|_{L^p} \leq \|A\|_{L^{p'}}^{\frac{1}{2}} \|A\|_{L^{p''}}^{\frac{1}{2}}$ $\frac{1}{2p} = \frac{1}{2p'} + \frac{1}{2p''} \Rightarrow p'' = n$

∴ $\|A \wedge A\|_{L^p} \leq \|A\|_{L^{2p}}^2 \leq \|A\|_{L^{p'}} \|A\|_{L^n} \leq C_2 \|A\|_{L^p} \|A\|_{L^n} \leq C_2 \eta \|A\|_{L^p}$
(elliptic estimate, with $d^*A=0$)

∴ $\|A\|_{L^p} \leq C_1 \|dA\|_{L^p} \leq C_1 (\|F_A\|_{L^p} + \|A \wedge A\|_{L^p}) \leq C_1 (\|F_A\|_{L^p} + C_2 \eta \|A\|_{L^p})$

Choose $\eta \leq \frac{1}{2C_1 C_2}$ $\|A\|_{L^p} \leq 2C_1 \|F_A\|_{L^p}$ (For $q \in P$, same, $C(q)$ continuous, $q \in [\frac{n}{2}, p]$ $\exists C_{\max}$)

$|(A \wedge A)(x)| = |A_i(x) A_j(x) \otimes dx^i \wedge dx^j|$, $|A_i A_j| \leq C |A_i| |A_j|$, $|A(x)|^2 = \sum g^{ij} |A_i(x)| |A_j(x)|$

$\|A \wedge A\|_{L^p}^p = (\int_M |A \wedge A|^p dV)^{\frac{1}{p}} \leq (\int_M C^p |A|^p dV)^{\frac{1}{p}} = C (\int_M |A|^p dV)^{\frac{1}{p}} = C \|A\|_{L^p}^p$
 $\|A\|_{L^p}^p = (\int_M |A|^p dV)^{\frac{1}{p}}$

Proof of Uhlenbeck's theorem:

For $A \in \mathcal{A}_\varepsilon$ (path connected component), $A_t \in \mathcal{A}_{1,\varepsilon}^P$ ($\|F_{A_t}\|_{L^{n/2}} \leq \varepsilon$) be a path with $A_0=0$, $A_1=A$. Let $S \subseteq [0,1]$ be the set of t s.t. $\exists \phi \in \mathcal{G}_2^P \subseteq L_2^P(\text{End } E)$, $d^* \phi(A_t)=0$, and $\|\phi(A_t)\|_{L^q} \leq N \|F_{A_t}\|_{L^2}$ for $\frac{n}{2} \leq q \leq p$, N is the constant in lemma above.
(U₁)
(U₂)

$0 \in S$, since A_0 is the $\begin{pmatrix} \text{flat connection in} \\ M \times \mathbb{R}^r, \text{ called gl or Dgl} \end{pmatrix}$ product connection, satisfies U_1, U_2 . $\Rightarrow S$ is nonempty.
[0,1] is connected, so we just need to prove S is both open and closed.

Openness: Take $t_0 \in S$, $\exists \phi_0$ s.t. $B = \phi_0(A_{t_0})$ satisfies:

(U₁) $d^* B = 0$; (U₂) $\|B\|_{L^q} \leq N \|F_B\|_{L^2}$

$$\|F_{\phi(A)}\|_0 = \|F_A \phi^{-1}\|_0 = \|F_A\|$$

By Sobolev embedding (in L^n)

$\|B\|_{L^n} \leq C \|B\|_{L^{n/2}} \leq CN \|F_B\|_{L^{n/2}} \leq CN \varepsilon$ ($B \in \mathcal{A}_{1,\varepsilon}^P$) (V₂)

So we could choose $\varepsilon < \frac{\eta}{CN}$, get: $\|B\|_{L^n} < \eta$ (V₃)

Now, consider $\phi_0(A_t) = B + b_t$, $b_{t_0} = 0$ (little change near t_0)

We want to find x , s.t. $\phi = \exp(x)$, $d^*(\phi(B + b_t)) = 0$.

Use linearization: at $x=0$, $b_t=0$

$$d^*(\phi(B + b_t)) \approx d^*(B + b_t) - D_B x + b_t$$

$$\begin{aligned} \phi(B + b_t) &= \phi(B + b_t) \phi^{-1} d\phi \phi^{-1} \\ &= (1 + \frac{1}{2} x^2 + \dots)(B + b_t)(1 - \frac{1}{2} x^2 + \dots) \approx (B + b_t) + [x, B + b_t] \\ d\phi &= d(1 + \frac{1}{2} x^2 + \dots) = x dx \\ \therefore \phi(B + b_t) &\approx B + b_t + [x, B] + [x, b_t] - dx \end{aligned}$$

$$\therefore d^*(B + b_t) \approx B + b_t - D_B x$$

$$\therefore d^* B = 0 \therefore d^*(\phi(B + b_t)) \approx d^*(B + b_t) - D_B x + b_t$$

Claim: ~~d^*D_B~~ d^*D_B is surjective to $(\ker d)^{\perp} = \text{Im } d^*$ (orthogonal) $(d^*D_B: L^p(g_E) \rightarrow \text{Im } d^* \subset L^p(g_E))$ ④

Pf: $d^*D_B x = d^*(dx + [B, x]) = d^*dx + d^*[B, x]$

(x is 0-form, $D_B x = dx + B \wedge x$, ~~$B \wedge x = B_i \cdot \sum \partial x^{i_1} \wedge \dots \wedge \partial x^{i_k}$~~ , $k=0 \Rightarrow B_i \cdot x$.)

$\therefore [B, x] = [B_i, x] \otimes dx^i$, here $B_i \cdot x$ is $\text{ad}_{B_i} x = [B_i, x]$, or B_i, x are matrix-valued fun, $B_i \cdot x = B_i x - x B_i$)

(since $(d^* \omega)_{i_2 \dots i_k} = -g^{ij} \nabla_j \omega_{i_2 \dots i_k}$, $[B, x]: 1\text{-form}$, $d^*[B, x]: 0\text{-form}$)
 $\nabla_j [B_i, x] = [\nabla_j B_i, x] - [B_i, \nabla_j x]$

$$d^*[B, x] = -g^{ij} \nabla_i ([B, x]_j) = -g^{ij} [\nabla_i B_j, x] - g^{ij} [B_j, \nabla_i x]$$

$$\therefore d^*B = -g^{ij} \nabla_i B_j \quad \therefore -g^{ij} [\nabla_i B_j, x] = [d^*B, x]$$

And $[B, dx] = g^{ij} [B_j, \nabla_i x]$ ($[B, dx] = [B_j, \nabla_i x] dx^j \wedge dx^i$, $\langle B, dx \rangle_g = g^{ij} [B_j, \nabla_i x]$)
 $[B, dx] = [B_j, \nabla_i x]$

$$\therefore d^*D_B x = d^*dx + [d^*B, x] - [B, dx]$$

$$\therefore d^*B = 0 \quad \therefore d^*D_B x = d^*dx - [B, dx]$$

(Hölder ineq for bi-linear operator)

$$\therefore \|d^*D_B x\|_{L^p} \geq \|d^*dx\|_{L^p} - \| [B, dx] \|_{L^p} \quad \text{and} \quad \| [B, dx] \|_{L^p} \leq \|B\|_{L^\infty} \|dx\|_{L^{p'}} \quad (\frac{1}{p'} = \frac{1}{p} - \frac{1}{n})$$

$$\geq \|d^*dx\|_{L^p} - \|B\|_{L^\infty} \|dx\|_{L^{p'}}$$

$$\therefore \text{By elliptic estimate, } \|dx\|_{L^p} \leq C_1 (\|dd^*x\|_{L^p} + \|d^*dx\|_{L^p}) = C_1 \|d^*dx\|_{L^p}$$

$$\text{and by Sobolev embedding, } C_2 \|dx\|_{L^p} \geq \|dx\|_{L^{p'}}$$

$$\therefore \|d^*D_B x\|_{L^p} \geq \frac{1}{C_1} \|dx\|_{L^p} - C_2 \|B\|_{L^\infty} \|dx\|_{L^p} \geq (\frac{1}{C_1} - C_2 \|B\|_{L^\infty}) \|dx\|_{L^p}$$

Choose η small enough s.t. $\frac{1}{C_1} - C_2 \eta > 0$, we get ~~d^*D_B~~ d^*D_B is surjective

★ $T = d^*D_B: L^p(g_E) \rightarrow L^p(g_E)$ $\|Tx\|_{L^p} \geq C \|x\|_{L^p}$ Consider $x_1, x_2 \in \ker d$ (if $x_1, x_2 \in \ker d$, $d^*D_B x = d^*dx - [B, dx] = 0 - 0 = 0$)
 $d(x_1 - x_2) = 0$, $x_1 - x_2 \in \ker d$, but we are in $\ker d$
 $\therefore x_1 - x_2 = 0$, T is injective // let $\{Tx_n\}$ be Cauchy seq, $\|x_m - x_n\|_{L^p} \leq \frac{1}{C} \|T(x_m - x_n)\|_{L^p} \rightarrow 0 \Rightarrow \{x_n\}$ is also a Cauchy seq. By completeness, $x_n \rightarrow x$, $Tx_n \rightarrow Tx$, $\text{Im } T$ is closed.

$T = d^*D_B$ is Fredholm operator, $L^p(g_E) = \text{Im}(T) \oplus \ker(T^*)$, $\dim \ker T = \dim \ker T^* \Rightarrow \ker T^* = 0$

$\therefore T$ surjective) (D_B is little disturb, $d^*D_B \sim d^*d$: self-adjoint, $\text{ind} := \dim \ker - \dim \text{coker} = 0 \Rightarrow \text{ind } d^*D_B = 0$)

From inj, $\dim \ker d^*D_B = 0 = \dim \text{coker } d^*D_B \therefore \text{coker } d^*D_B = L^p(g_E) / \text{Im } d^*D_B \rightarrow \text{ran } d^*D_B \text{ is full space!})$

Then, $(x, b_t) \rightarrow d^*(\exp(x)(B+b_t))$ is smooth, at $(0,0)$, linearization is onto
 then at t_0 , has solution $x=0$.

So by implicit function thm, $\exists$ neighborhood of t_0 , smooth family x_t , s.t.

$$d^*(\exp(x_t)(B+b_t))=0$$

def $G_t = \exp(x_t) \cdot G_0$, then ~~G_t~~ satisfy U1.
 $G_t(A_t)$

and:

At t_0 , $\|G_{t_0}(A_{t_0})\|_{L^1} = \|B\|_{L^1} < \eta$, ~~$G_t(A_t)$ depends smoothly on t .~~ it is an open condition

$\therefore$ near to, there is still $\|G_t(A_t)\|_{L^1} < \eta$.

$\therefore$ By lemma, $\exists N$, s.t. $\|G_t(A_t)\|_{L^1} \leq N \|F_{A_t}\|_{L^2} \quad (\forall \frac{n}{2} \leq q \leq p)$

$\therefore U_2$.

$\therefore S$ is open. ($\forall t_0, \exists$ neighbor of t_0 , in $S \Rightarrow S$ is open).

Closedness: Let $t_i \in S, t_i \rightarrow t_\infty \in [0,1]$

Def $A_i = A_{t_i} \rightarrow A_{t_\infty} = A_\infty$ in $\mathcal{A}^p$

we know that $\exists \delta_i \in \mathbb{R}^p$ s.t. $A'_i = G_i(A_i)$ satisfies:

$$U1: d^*A'_i = 0; \quad U2: \|A'_i\|_{L^1} \leq N \|F_{A_i}\|_{L^2} \quad (\forall \frac{n}{2} \leq q \leq p)$$

By U_2 , $\{A'_i\}$ is bounded in L^p_1 . (let $q=p, t: [0,1] \rightarrow A_t \rightarrow \|F_{A_t}\|_{L^p}$ continuous on cpt set $[0,1]$)

$\therefore \|F_{A_t}\|_{L^p} \leq M \quad \therefore \|A'_i\|_{L^p} \leq M$ so by Banach-Alaoglu thm, (closed unit ball is weak cpt, i.e.

any bounded sequence has ~~converge~~ weak converge subsequence) $\exists$ subsequence, for convenience,

let them still be A'_i , s.t. $A'_i \rightarrow A'_\infty$

Then, $A'_i = G_i A_i G_i^{-1} - dG_i G_i^{-1}$ $\therefore dG_i = G_i A_i - A'_i G_i$ (Bootstrap)

$\therefore G_i \in L^\infty$ ($G(x): E_x \rightarrow E_x$ is linear iso $\rho(x) \in G, G$ is cpt, $G \hookrightarrow \mathbb{R}^2 \therefore \forall g \in G, |g| \leq C. \|G_i\|_{L^\infty} = \text{esssup } |G_i(x)| \leq C$)

$$A_i, A'_i \in L^p_1 \subset L^{p'} \quad (p' = \frac{np}{n-p})$$

weak converge: $\{x_n\} \rightarrow x$ if $\forall f \in X^*$
 $f(x_n) \rightarrow f(x)$

$\therefore dG_i$ is bounded in $L^{p'}$, G_i is bounded in $L^{p'}$

$\exists$ subsequence $G_i \rightarrow G_\infty$ in $L^{p'}$

what we want: $A_\infty \in S$
 $\Leftrightarrow A'_i \rightarrow A'_\infty$ also satisfy
 U_1, U_2 . But this is
 strong converge, it may not
 true especially $G_i(A_i)$ is not
 linear. So we can only use
 weak converge, try to
 get $A_\infty = G_\infty(A_\infty)$ indeed.
 (Indeed for $A_\infty, \exists G_\infty$ s.t. $G_\infty(A_\infty) = G_i(A_i)$)

$\nabla dG_i = \nabla G_i \cdot A_i + G_i \nabla A_i - \nabla A_i \cdot G_i - A_i' \nabla(G_i^{-1})$ (6)
 $\nabla G_i \in L^{p'}, A_i \in L^{p'} \Rightarrow \nabla G_i \cdot A_i$ is bounded; $G_i \in L^\infty, \nabla A_i \in L^{p'} \Rightarrow G_i \nabla A_i$ is bounded (in $L^{p'}$)
 $\nabla A_i \in L^{p'} \Rightarrow \nabla A_i \cdot G_i$ is bounded; $A_i' \in L^{p'}, \nabla(G_i^{-1}) \in L^{p'} \Rightarrow A_i' \nabla(G_i^{-1})$ is bounded ($p' = \frac{np}{n-p} > 2p$)

$\therefore \nabla dG_i$ is bounded in $L^{p'}$ ($L^{p'} \times L^{p'} \rightarrow L^{p'}$ is continuous)
 $\therefore G_i$ is bounded in L_2^p (In fact, $\frac{1}{2p} + \frac{1}{2p} = \frac{1}{p}$, by Hölder, $\|fg\|_{L^p} \leq \|f\|_{L^{2p}} \|g\|_{L^{2p}}$)
 $\therefore \exists G_i$ (subsequence) $\rightarrow G_\infty$ in L_2^p (cpt ∇ $\nabla G_i \rightarrow \nabla G_\infty$ in $L^{p'}$)

Now, $G_i \rightarrow G_\infty$ in $L_2^p \therefore dG_i \rightarrow dG_\infty$ in L_1^p ; $G_i A_i \rightarrow G_\infty A_\infty$ in L_1^p (multiply in weak topo)
 $A_i' G_i \rightarrow A_\infty' G_\infty$ in L_1^p (weakly converge is unique) (Also, $A_i \rightarrow A_\infty$ in L_1^p)
 $\therefore dG_\infty = G_\infty A_\infty - A_\infty' G_\infty \Rightarrow A_\infty' = G_\infty(A_\infty)$

Then, check that: (As limit, A_∞ has Coulomb gauge)
 $d^*: L_1^p \rightarrow L^p$ is continuous linear operator $\therefore d^* A_\infty' = \lim d^* A_i = 0 \therefore U_1 \checkmark$

By Fatou lemma: $\int_S \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int_S f_n \, d\mu$ (6)
 $\therefore \liminf_{i \rightarrow \infty} \|A_i'\|_{L_1^p} \leq N \liminf_{i \rightarrow \infty} \|F_{A_i}\|_{L^p} = N \|F_{A_\infty}\|_{L^p}$ ($A_i \rightarrow A_\infty$ strongly)

$(|F_{A_\infty}| = |G_\infty F_{A_\infty} G_\infty^{-1}| = |F_{G_\infty(A_\infty)}|$ almost everywhere.)

$\therefore \|F_{A_\infty}\|_{L^p} = \|F_{G_\infty(A_\infty)}\|_{L^p}$

$\therefore \|A_\infty'\|_{L_1^p} \leq N \|F_{G_\infty(A_\infty)}\|_{L^p} \therefore U_2 \checkmark$

$\therefore t_\infty \in S$, S is closed

If $f_i \rightarrow f$ in L_2^p , $g_i \rightarrow g$ in L_1^p , $p > \frac{n}{2}$, then $f_i g_i \rightarrow f g$ in L_1^p
 Pf: $f_i g_i - f g = (f_i - f) g_i + f(g_i - g)$, for $h \in (L_1^p)^*$
 Consider $\langle f_i g_i - f g, h \rangle: L_2^p \hookrightarrow C^0$, $f_i \rightarrow f$ in C^0 , $\|f_i - f\|_{C^0} \leq \|f_i - f\|_{L_2^p} \|g_i\|_{L_1^p} \rightarrow 0$
 $\| \nabla(f_i - f) g_i \|_{L^p} \leq \|(\nabla f_i - \nabla f) g_i\|_{L^p} + \|f_i - f\|_{C^0} \|g_i\|_{L_1^p} \rightarrow 0$
 $L_1^p \hookrightarrow L^\infty$, $\nabla f_i \rightarrow \nabla f$ in L^∞ , $g_i \in L_1^p \hookrightarrow L^1$: $\|(\nabla f_i - \nabla f) g_i\|_{L^p} \leq \|\nabla f_i - \nabla f\|_{L^\infty} \|g_i\|_{L^1} \rightarrow 0$
 $\therefore \|(\nabla f_i - \nabla f) g_i\|_{L^p} \leq \|\nabla f_i - \nabla f\|_{L^\infty} \|g_i\|_{L^1} \rightarrow 0$ ($L_1^p \hookrightarrow L^1$, $\nabla f_i \rightarrow \nabla f$ in L^∞)
 $\therefore \langle f_i g_i - f g, h \rangle \rightarrow 0$ in L_1^p strongly
 $h \in (L_1^p)^* \rightarrow \langle f_i g_i - f g, h \rangle = \langle g_i - g, f h \rangle$ $f \in L_2^p$, $\forall \psi \in L_1^p$, $L_2^p \times L_1^p \rightarrow L_1^p$ continuous
 $\therefore f \psi \in L_1^p \therefore \langle h, f \psi \rangle \leq \|h\|_{L_1^p} \|f \psi\|_{L_1^p} \leq C \|h\|_{L_1^p} \|f\|_{L_2^p} \|\psi\|_{L_1^p}$
 $\therefore \psi \rightarrow \langle f h, \psi \rangle$ is linearly bounded on L_1^p , so $f h \in L_1^p$ or $(L_1^p)^*$
 $\therefore f h \in L_1^p \therefore \langle g_i - g, f h \rangle \rightarrow 0$